

Predicting Expected Time of Arrival of Shipments Through Multiple Linear Regression



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Abstract Handling uncertainty in information is one of the biggest challenges faced by logistics management and freight forwarding agencies. The uncertainty may be in user needs, availability of raw materials, Estimated Time of Arrivals (ETA) of shipments, or occurrence of any natural calamities. These agencies are increasingly adopting intelligent ways of handling uncertainties in the information. In this paper, we describe a machine learning-based approach to deal with the uncertainty in the Estimated Time of Arrivals (ETA) of shipments. The approach uses multiple regression techniques to predict the ETA of a shipment at the destination port. The approach uses data from two different sources. First is the in-house internal data about shipments maintained by a company. Second is the marine traffic data collected through the Automatic Identification System (AIS). The approach combines these two data to predict the ETA of a shipment. The approach predicts the ETA of the shipment with 89% accuracy. The prediction model described in the paper is being in use and the predicted ETA has been found helpful to effectively manage resources in the downstream supply chain such as human sources at port, transporter and customers inquiries.

Keywords Automatic identification system · Marine transportation · Predictive analytics · Machine learning · Logistics

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1 Introduction

Knowing information about the Expected Time of Arrival (ETA) of a shipment is critical for logistics management and freight forwarding agencies [2]. Estimates about ETA are commonly used to answer queries of customers, to issue contracts to liners, to allocate human resources at ports, to manage warehouses, and overall to run the business smoothly. Accurate estimates of ETA can also improve resource utilization and reduce operating costs.

Estimating ETA is one of the extensively researched problems in transportation and supply chain industry [4–6]. However, for logistics management and freight forwarding agencies, the techniques and models developed to estimate ETA in a general context are often inadequate. Such models fail to utilize organization-specific information and knowledge gathered over the years, dealing with specific liners or carriers. Hence, the estimates of ETA that merely rely either on historic or streamed data of traffic fail to capture the operating behaviour of carriers or liners that these agencies deal with regularly.

Though the task of estimating ETA is crucial for logistics management and freight forwarding agencies, it poses numerous challenges to arrive at an acceptable value of ETA [7]. First, the majority of the logistic business rely on verbal communication. Second, logistics and freight forwarding business is enacted through various contractors and subcontractors. Third, the information required to predict ETA is often scattered and present in unstructured documents.

In this paper, we present a solution for estimating ETA of shipments specific for the agencies involved in the business of freight forwarding that prefers marine transportation to transfer goods. The approach uses two different data sources. First is the data about maritime traffic acquired through the Automatic Identification System (AIS) [8]. Second is in-house company data that capture the operating behaviour of liners or carriers. We use the multiple regression analysis techniques to predict the value of ETA of a shipment at a port.

2 Approach

The conceptual diagram in Fig. 1 captures the main elements of our approach. The purpose of this diagram is to make explicit the tacit knowledge assumed in various business processes carried out by logistics and freight forwarding agencies. It states that:

1. Vessels carry Shipments.
2. The position of a vessel is defined by Automatic Identification System (AIS). The AIS is a self-reporting navigational system mandatory to be installed onboard every ship. More information about AIS can be found in [1].
3. An AIS data provider can be a commercial or a non-profit entity which supplies AIS data. Logistic management or a freight forwarding agency need to subscribe to

an AIS data provider to have historical or up-to-date AIS data. The *MarineTraffic* (marinetraffic.com) is one of the commercial AIS data providers.

4. A vessel is identified by AIS data attributes. Each ship has a unique vessel name. The latitude and longitude of a location define the current position of a boat in the ocean. When a vessel transmits AIS messages, the message is timestamped with *FileTimeStamp*. The *createdTimeStamp* is the time when an AIS data provider records the received AIS message in the system.
5. Logistic management and freight forwarding agencies internally maintain shipment data. A shipment is defined by attributes such as a unique shipment number, the port-of-loading (POL), the port-of-delivery (POD), a carrier, a vessel name, the actual time of departure and arrival. The flag *isTransshipped* denotes whether the shipment is a multi-hop shipment. A sample of shipment data is shown in Table 1.
6. The fused data is the merged data consisting of external of AIS data and internal shipment data agreeing on a common vessel name.

The fused data is then processed to make it ready to input format for building a multiple regression model.

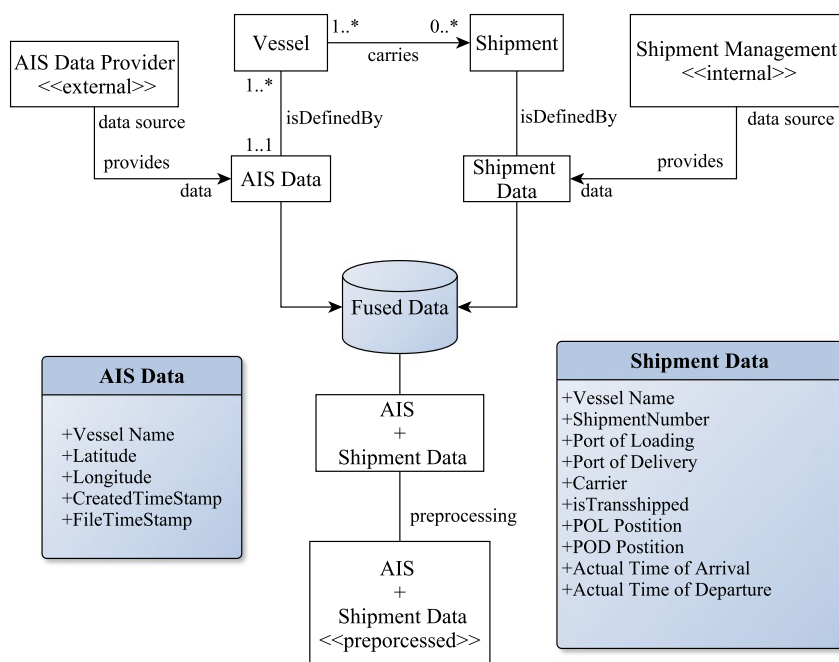


Fig. 1 Main elements in our approach

Table 1 A sample of shipment data

Vessel name	Ship No.	POL	POD	Carrier	IsTS	ATA	ATD	POL Cord	POD Cord.
ABC	X002	HAMB	IZMIR	Hapag-Lloyd	Yes	10-05-2016	16-04-2016	53.55, 9.99	38.41, 27.03
PQR	X790	MUND	HAMB	Hapag-Lloyd	No	05-05-2016	14-04-2016	22.75, 69.70	53.54, 9.99
XYZ	X211	MUND	BARC	UASC	No	17-05-2016	28-04-2016	22.75, 69.70	41.38, 2.17
LML	X376	MUND	AMBA	Safma	Yes	25-04-2017	09-04-2017	22.75, 69.70	40.96, 28.67
SQR	X017	AMBA	JEDD	UASC	No	22-05-2017	27-04-2017	40.96, 28.67	21.57, 39.15

3 Model Development

This section describes the steps adopted to build the prediction model using multiple regression technique [3].

3.1 Data Pre-processing

The fused data consist of data from two sources. The first one is external AIS data which reflects the vessel's current position in the ocean. The AIS periodically transmits vessel information. As shown in Table 2, an AIS message contains the longitude and latitude of the vessel's current position and not the distance travelled by vessel. The first objective of the preprocessing step is to calculate the distance travelled by a vessel from the port-of-loading (POL) when an AIS provider receives the AIS message from the vessel. We used two latest AIS messages received from the vessel of interest for model building. We refer the corresponding distances travelled by ship as $L1$ and $L2$. The $L1$ is the most recent distance, and $L2$ is the second

Table 2 A sample of AIS data

Name	Latitude	Longitude	CreatedTS	FileTS
ABC	21.95	68.00	17-04-2016	16-04-2016
PQR	22.63	69.17	15-04-2016	14-04-2016
XYZ	19.71	70.36	28-04-2016	28-04-2016
LML	14.20	52.11	14-04-2017	14-04-2017
SQR	21.80	68.93	27-04-2017	27-04-2017

Table 3 Calculating distance covered

```

calculate_distance<-function(long1, lat1, long2, lat2){
rad <- pi/180

#convert longitude and latitude to radian

lat1_rad <- lat1*rad
long1_rad <- long1*rad
lat2_rad <- lat2*rad
long2_rad <- long2*rad

#calculate distance between latitudes and longitudes
dlon <- long2_rad - long1_rad
dlat <- lat2_rad - lat1_rad

#Haversine formula

a <- (sin(dlat/2))^2 + cos(a1)*cos(b1)*(sin(dlon/2))^2
c <- 2*atan2(sqrt(a), sqrt(1 - a))
R<- 6378137 #Radius of the earth
d<- R*c #final distance
return(d)
}

for (everyVessel in dataset){
  for (everylocation for everyVessel){
    L=calculate_distance(long1, lat1, long2, lat2)
  }
}

```

most latest distance covered by the vessel of interest. The method implemented in R programming language to calculate the distance travelled by a vessel from the given longitude and latitude is given in Table 3. The second objective of the preprocessing step is to associate shipment data and vessel data that agree on the common value of vessel name and project out the attribute such as POL, POD, carrier and ATA from the shipment date. The sample of processed data is shown in Table 4.

3.2 Data Analysis

We started the data exploration and analysis activity with the intuitive idea that expected time of arrival of a shipment depends on the distance travelled by a shipment so far, port-of-landing, port-of-delivery, and carrier who owns the vessel. To validate our intuition, we checked the strength of correlation between input variables

Table 4 A sample of Data after pre-processing

POL	POD	Carrier	L1	L2	ATA	Predicted ETA
1	2	1	28.39	34.07	20-05-2016	20-05-2016
2	1	2	7.46	8.95	30-06-2016	01-07-2016
3	2	1	8.10	9.72	01-06-2017	30-05-2017
1	3	2	17.04	20.45	12-02-2018	13-02-2018
2	1	2	15.62	18.74	03-03-2017	03-03-2017

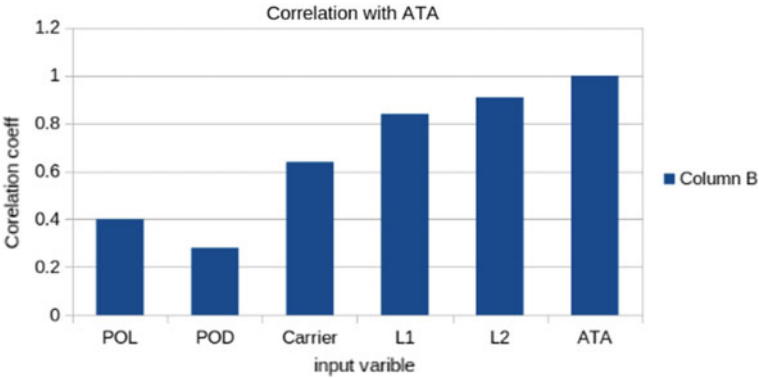


Fig. 2 Correlation of input variables with ATA

and the output variable, i.e. ATA. In our case, the input variables are POL, POD, Carrier, *L1* and *L2*. As defined in the previous section, *L1* and *L2* are the distances travelled by vessel as reported through two latest messages. Figure 2 shows the correlation between input variables and ATA. The highest correlation is observed between *L2* (0.91) and ATA, while the lowest correlation is observed between POD(0.28) and ATA. The other input variables, i.e. *L1* (0.84), carrier (0.64), and POL(0.4) also demonstrate a positive correlation with ATA. Hence, we selected all five input variables to build the multiple linear regression model..

3.3 Multiple Regression Model

The multiple regression analysis is predication technique that identifies a relationship among input and output variables. The general format for the multiple regression technique is:

$$y = \beta_0 + \beta_1xX_1 + \beta_2xX_2 + \cdots + \beta_nxX_n$$

where *y*: is output variable, in our case it is ATA

Table 5 Regression coefficients

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.1823593	0.030148	31.103	<2e−16	***
POL	0.0035776	0.0006051	5.913	3.39e−09	***
POD	0.0012442	0.0002230	5.584	2.36e−08	***
Carrier	0.0076308	0.0013385	5.701	1.20e−08	***
L1	0.0676596	0.0044488	15.209	<2e−16	***
L2	1.1038425	0.0040762	270.802	<2e−16	***
Signif. codes	0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1				
Residual standard error	3.69 on 53,012 degrees of freedom				
Multiple R squared	0.9324	Adjusted R squared		0.9323	
F-statistic	1.461e+05 on 5 and 53,012 DF		p-value	<2.2e−16	

$X_1, X_1, \dots, X_n$ are input variable, in our case POL, POD, carrier, L1 and L2
 $\beta_0, \beta_1, \dots, \beta_n$ are regression coefficients.

The purpose of model building is to find the values of $\beta_0, \beta_1, \dots, \beta_n$ from the training data sets.

We have used the method 'lm' from the R programming language to build the multiple regression model. The model parameters obtained when the model is trained on 53,018 data items are shown in Table 5.

3.4 Model Evaluation

We used the summary information shown in Table 5 and obtained through *summary()* command in R. This information tells us how the model fits our data and how dependent and independent variables are related to each other. The most important information is in the last two columns of the Table, i.e. *p-value* and *significance level*. A lower *p-value* indicates that independent variables are statistically significant. Generally, the threshold used for *p-value* is 5%. In our case, the *p-value* is below the threshold for all the independent variables, so they are statistically significant. The Multiple R-squared value is a measure of how well our model as a whole performs or in other words, how t strongly the dependent variables and independent variables are related. The closer the value to 1.0, the better is the model. Since the R-Squared value is 0.9324, we know that model explains nearly 0.93% of the variations in the dependent variables.

4 Conclusion

Accurately predicting ETA of a shipment is a challenging and significant activity for the overall functioning of logistics and supply chain management organizations. This paper describes a novel approach of combining data external to an organization and internal organization-specific data to predict the expected arrival time of a shipment. The paper describes the approach based on Multiple Linear Regression and observes that port of loading, the port of delivery, intermediate distance travelled by vessel, and vessel operator are statistically significant parameters to predict ETA of a shipment. The model reported in this paper predicts the ETA with an acceptable degree of error. It will be interesting to explore the performance of other machine learning techniques such as Support Vector Machines and Neural Network for the task of predicting ETA.

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